

Lecture 27: Alternative Water Reactors (BWR & CANDU)

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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1 Introduction

While the Pressurized Water Reactor (PWR) utilizes a single-phase coolant loop and a secondary steam cycle, alternative designs have captured significant market share by simplifying the loop (BWR) or utilizing different moderators to enable natural uranium fuel (CANDU). This lecture covers the physics and engineering trade-offs of these designs.

2 The Heavy Water Reactor (CANDU)

The CANDU (CANadian Deuterium Uranium) design prioritizes **Neutron Economy** to allow the use of unenriched (Natural) Uranium ($0.7\% \text{ }^{235}\text{U}$). It shares some features with a PWR (pressurized coolant, water moderator), but the use of heavy water and a pressure-tube architecture leads to fundamentally different neutronics and mechanical design.

2.1 Global Prevalence and History

Currently, there are **31 official CANDU reactors** operating worldwide, representing about 7% of the global nuclear fleet.

- **Canada (The Primary Fleet):** Canada operates the vast majority (19 units), heavily concentrated in Ontario, where they provide over 50% of the province's electricity grid.
- **The Export Market:** Canada has successfully exported genuine CANDU units to five countries: South Korea, Romania, China, Argentina, and Pakistan.
- **The Indian Fleet (IPHWRs):** India is actually the second-largest operator of this technology, though they do not use the "CANDU" name. After India used byproducts from a Canadian-supplied research reactor for their 1974 nuclear weapons test ("Smiling Buddha"), Canada abruptly cut off all nuclear trade. Forced to proceed independently, India reverse-engineered the design to build their own domestic fleet of approximately 18 Indian Pressurized Heavy Water Reactors (IPHWRs). If these unlicensed derivatives are included, the global footprint of heavy-water, pressure-tube reactors is closer to **50 units** ($\approx 11\%$ of the global fleet).

2.2 Physics: The Heavy Water Trade-off

CANDU uses Deuterium Oxide (D_2O) as the moderator.

- **Absorption Advantage:** $\sigma_a^H = 0.33$ barns vs. $\sigma_a^D = 0.0005$ barns. Deuterium captures almost no neutrons, preserving enough neutron population to maintain criticality with natural uranium.
- **Slowing Down Disadvantage (Quantitative):** Deuterium is twice as heavy as Hydrogen, making it less efficient at removing kinetic energy per collision.
 - Logarithmic Energy Decrement: $\xi_{H_2O} \approx 0.92$ vs. $\xi_{D_2O} \approx 0.51$.
 - Collisions to thermalize: Light Water ≈ 19 ; Heavy Water ≈ 35 .
 - *Consequence:* Neutrons travel much further while slowing down. The lattice pitch must be large ($\approx 28 - 30$ cm) compared to a PWR fuel rod spacing of (≈ 1.2 cm).

2.3 Design: The Pressure Tube Concept

Because the lattice is so large, a single pressure vessel would be prohibitively large and thick.

- **Calandria:** The moderator is kept in a large, low-pressure tank called the Calandria.
- **Pressure Tubes:** The fuel and coolant run through hundreds of individual horizontal pressure tubes that pierce the tank.
- **On-Line Refueling:** Because natural uranium has low excess reactivity, fuel must be replaced constantly. Robots clamp onto the pressure tubes while the reactor is running to push new bundles in and old bundles out.

2.3.1 Where Does the Moderation Happen? (The Two-Fluid System)

A common misconception is that the heavy water flowing over the tightly packed fuel pins acts as the moderator. While some moderation occurs within the pressure tube, the volume and geometry are insufficient to fully thermalize neutrons efficiently. Most moderation occurs in the large surrounding calandria volume.. Unlike a PWR, the CANDU physically separates the coolant and the moderator into two isolated systems:

- **The Coolant (Inside the Tubes):** The D_2O flowing through the pressure tubes is hot ($\approx 310^\circ\text{C}$), highly pressurized, and acts purely as a heat transfer fluid. Fast neutrons born in the fuel bundle easily escape the tube with very little slowing down.
- **The Moderator (Outside the Tubes):** The actual moderation occurs in the **Calandria** tank. The Calandria is filled with a separate, massive volume of unpressurized, cold D_2O ($\approx 70^\circ\text{C}$).

The Neutron Journey & Lattice Pitch: Because D_2O requires roughly 35 collisions to thermalize a neutron, the pressure tubes must be spaced far apart (a lattice pitch of ≈ 28.6 cm). A fast neutron is born in one fuel bundle, escapes the pressure tube, bounces around in the cold D_2O of the Calandria until it thermalizes, and then drifts into a *neighboring* pressure tube to initiate a new fission. This separation allows the moderator to remain cold and dense (maximizing neutron economy) while the coolant runs hot (maximizing thermodynamic efficiency).

2.3.2 The Fuel Channel Geometry

To prevent the 310°C pressure tube from boiling the 70°C moderator, the fuel channel utilizes a "tube-within-a-tube" design:

1. **Inner Pressure Tube:** A thick Zirconium-Niobium pipe holding the fuel bundles and high-pressure coolant.
2. **Outer Calandria Tube:** A thinner Zircaloy pipe welded directly to the Calandria tank walls.
3. **Annulus Gas:** The gap between the two tubes is filled with a continuously circulating insulating gas (like CO_2) to prevent heat loss to the moderator. Garter springs keep the inner tube from sagging and touching the outer tube.

2.3.3 The Fuel Bundle and Thermal Flux Depression

Unlike the massive, full-core-height fuel assemblies of a PWR, CANDU fuel is packaged in short, compact bundles. A standard bundle is roughly 50 cm long, weighs 24 kg, and has an outer diameter of approximately **10.2 cm**.

- **Internal Geometry:** Inside this 10.2 cm diameter, 37 individual fuel elements are arranged in concentric rings (1 center pin, surrounded by rings of 6, 12, and 18).
- **Tight Tolerances:** The pins are relatively thick (≈ 13.1 mm in diameter). The 18 pins on the outer ring are packed so tightly that the physical gap between them is roughly **1 to 2 millimeters**.
- **Spatial Self-Shielding (Flux Depression):** Because the outer ring forms a dense barrier of uranium, thermal neutrons diffusing in from the bulk moderator are heavily absorbed by the outer ring. The outer pins physically "shield" the inner pins.
- **Operational Consequence:** The thermal flux at the center of the bundle is 20% to 30% lower than at the outer edge. This limits the total power of the reactor, as operators must keep the hottest outer pins below their thermal limits (melting/dryout), leaving the inner pins operating well below their potential capacity.
- **Advanced Designs:** To combat this, newer bundles (like the CANFLEX 43-element design) use thinner pins on the outside to allow more neutron penetration, and thicker pins in the center, successfully flattening the power profile.

2.4 Economics: The Cost of Heavy Water

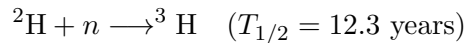
The single largest economic hurdle for the CANDU design is the staggering cost of the heavy water itself. Because deuterium is naturally present in only about 1 in 6,400 water molecules, extracting it requires massive, energy-intensive chemical separation plants.

- **Unit Cost:** Industrial bulk nuclear-grade heavy water typically costs between \$300 to \$800 per kilogram.

- **Capital Penalty (The "Fill-Up"):** A standard commercial CANDU requires roughly **500 metric tons** (500,000 kg) of heavy water to fill both the Calandria and the primary coolant loops. This means the heavy water alone adds approximately **\$150 million to \$400 million** to the upfront capital cost of building the plant.
- **The Capital vs. Fuel Trade-off:** The economics of a CANDU balance out over its 60-year lifespan. While a PWR gets its light water essentially for free, it locks the utility into paying millions annually for uranium enrichment. The CANDU pays a massive upfront capital penalty for the moderator, but benefits from incredibly cheap, unenriched natural uranium fuel for the life of the reactor.
- **Leak Recovery:** Because the fluid is essentially liquid gold, CANDU plants are meticulously engineered with vapor recovery systems (often utilizing cold silica gel beds) to catch, condense, and recycle nearly every drop of heavy water that evaporates from pump seals or valves.

2.5 Issues: Tritium & Efficiency

1. **Tritium Production & Economics:** Deuterium absorbs neutrons to become Tritium (^3H):



- **The Hazard (Safety First):** Inside the reactor, neutron capture turns the D_2O into DTO (Deuterium Tritium Oxide). If this leaks into the plant atmosphere, rapid isotopic exchange with normal humidity converts it into HTO (Tritiated Water). Because it is chemically identical to water, HTO is a significant ingestion and skin-absorption hazard for workers. The primary reason utilities build massive detritiation facilities is to reduce this occupational radiation exposure, not purely to make a profit.
 - **The Production Rate:** A standard CANDU produces approximately **130 grams** of Tritium per year as the heavy water absorbs neutrons.
 - **The Income Stream:** Because global supply is incredibly tight, Tritium is highly valuable ($\approx \$30,000/\text{gram}$). A single reactor generates roughly **\$3.9 million** worth of Tritium annually. Centralized facilities (like the Darlington Tritium Recovery Facility in Ontario) harvest roughly 2 to 2.5 kg per year from the fleet, yielding up to **\$75 million** annually. This revenue helps offset the massive capital costs of the D_2O and the detritiation plant itself.
 - **The Market:** Once harvested, it is sold for:
 - Self-powered emergency lighting (exit signs, watch dials).
 - Medical and Life-science tracers.
 - Fusion Research (e.g., the ITER project).
 - Nuclear Weapons (Boost gas).
2. **Thermodynamic Efficiency:** The pressure-tube design and material limits constrain outlet temperatures to lower values than a light water PWR.
 - *Limitations:* The outlet temperature is restricted to $\approx 310^\circ\text{C}$.
 - *Result:* The thermodynamic efficiency is lower, typically in the range of 28% to 30% (compared to $\approx 33\%$ for a PWR).

- *Solutions:* One approach is to use other fluids in the coolant loop such as light water, but that requires at least some Uranium enrichment due to neutron absorption.
3. **Fuel Utilization vs. Burnup:** There is a critical distinction between *burnup* and *resource utilization*.

- **Low Burnup:** Because CANDU fuel is natural uranium (0.7% ^{235}U), it can only sustain criticality for a short time. The discharge burnup is low ($\approx 7,500 \text{ MWd/MTU}$) compared to a PWR using enriched fuel ($\approx 50,000 \text{ MWd/MTU}$).
- **High Waste Volume:** This means a CANDU produces $\approx 7\times$ the *volume* of spent fuel bundles compared to a PWR for the same energy generated.
- **High Resource Efficiency:** However, because a PWR requires $\approx 8 - 10 \text{ kg}$ of mined uranium to produce 1 kg of enriched fuel (discarding the rest as tailings), the CANDU actually extracts *more energy per kg of mined uranium* than a PWR.

The Mass Balance Problem: Where does the ^{235}U go?

To understand why the CANDU has higher overall resource efficiency despite its low fuel burnup, we must look at the actual ^{235}U concentrations throughout the fuel cycle:

- **Natural Uranium Ore:** $\approx 0.71\%$
- **Enrichment Tailings (Depleted U):** $\approx 0.25\%$
- **Spent CANDU Fuel:** $\approx 0.2\%$
- **Spent PWR Fuel:** $\approx 0.8\%$ to 1.0%

The PWR Trap: To make 1 kg of 4% enriched PWR fuel, you must mine roughly 8 to 10 kg of natural uranium ore. The enrichment plant throws away the excess mass as "tails" containing roughly 0.25% ^{235}U . The reactor then burns the enriched fuel, but discharges it when it still has roughly 0.8% ^{235}U remaining (ironically, richer than the natural ore pulled out of the ground!). In a PWR cycle, you lose ^{235}U in two separate places.

The CANDU Advantage: The CANDU skips the enrichment process entirely (zero enrichment tails) and burns the natural uranium straight down to $\approx 0.2\%$ before discharge. Because it avoids enrichment losses and also benefits from breeding and burning ^{239}Pu along the way, the CANDU extracts about 15% to 20% more total thermal energy from the original ton of mined ore than a PWR.

The CANDU Waste Disadvantage: While the total mass of "leftovers" is similar for both cycles, the *radiological hazard* is vastly different. For every 10 kg of mined uranium, a PWR produces roughly 1 kg of highly radioactive High-Level Waste (spent fuel) and 9 kg of Depleted Uranium (DU) tailings. DU is only mildly radioactive, safe to handle, and has industrial/military uses (e.g., kinetic energy penetrators, radiation shielding). In contrast, the CANDU puts all 10 kg of that natural uranium through the core, meaning **100% of the initial mass becomes HLW** contaminated with fission products, requiring complex shielding, cooling, and ultimate deep geological disposal.

2.6 CANDU Safety and Containment Strategy

By eliminating the massive, ultra-thick steel pressure vessel used in PWRs and BWRs, the CANDU design fundamentally alters its safety profile and containment requirements.

1. **Distributed Pressure and Passive Heat Sinks:** The high-pressure coolant is distributed across roughly 380 to 480 individual channels rather than a single massive volume.
 - *Small Break LOCA:* A single pressure tube rupture only depressurizes one channel. Emergency core cooling systems can easily compensate, preventing a core-wide transient.
 - *The Calandria as a Heat Sink:* If a channel completely dries out, the overheating inner pressure tube will physically sag and touch the cooler outer calandria tube. The massive volume of 70°C heavy water in the calandria acts as a passive heat sink, drawing decay heat away from the fuel and significantly delaying or preventing a localized meltdown.
2. **The "Spaghetti" Problem and Header Breaks:** The trade-off for eliminating the large pressure vessel is the creation of thousands of individual mechanical connections.
 - *Feeder Pipes:* The face of the reactor is covered in a dense, complex web of feeder pipes that route coolant into and out of the individual channels.
 - *Large Break LOCA:* These tiny pipes connect to large distributor pipes called "headers." A rupture of a main header represents the CANDU equivalent of a Large Break LOCA, depressurizing a significant fraction of the core simultaneously.
 - *Positive Void Coefficient:* Unlike Light Water Reactors, a CANDU has a slightly positive void coefficient of reactivity. If a header breaks and coolant flashes to steam (voids), the formation of steam voids alters moderation and neutron leakage in a way that produces a net positive reactivity insertion. Consequently, CANDUs are engineered with **two fully independent, ultra-fast shutdown systems** (spring-assisted solid shutoff rods dropping from the top, and high-pressure liquid gadolinium poison injected from the side).
3. **The Vacuum Building Containment Solution:** Because a header break would still fill the reactor building with radioactive steam, CANDUs require robust containment. However, Canadian plants are typically built as massive multi-unit stations (e.g., the Bruce Nuclear Generating Station operates 8 reactors on one site).
 - Instead of building an ultra-high-pressure dome for every single reactor, the individual reactor buildings are kept at a slightly negative pressure and connected via massive relief valves to a central, shared **Vacuum Building**.
 - This massive concrete silo is maintained at a near-absolute vacuum. In the event of a LOCA in any reactor, the overpressure blows the valves open, instantly sucking the radioactive steam out of the reactor building. A dousing tank in the roof of the vacuum building sprays cold water to condense the steam and drop the pressure of the entire complex to safe levels within seconds.

Historical Case Study: The 1983 Pickering Tube Rupture

The theoretical safety advantages of the CANDU's distributed pressure system were put to a real-world test on August 1, 1983, at the Pickering Nuclear Generating Station in Ontario.

The Failure Mechanism: While operating at full power, a 2-meter-long crack suddenly unzipped along pressure tube G16 in Unit 2. The root cause was traced back to the **garter springs** designed to separate the inner pressure tube from the outer calandria tube. A spring had shifted out of place, allowing the heavy, 310°C inner tube to sag and physically contact the cold (70°C) calandria tube. This extreme temperature gradient caused hydrogen and deuterium isotopes in the metal to migrate to the cold contact point, creating brittle **hydride blisters**. The embrittled Zircaloy-2 metal ultimately shattered under the 9 MPa of coolant pressure.

The Safety Case Proven: If a massive PWR pressure vessel had unzipped in this manner, it would have been a catastrophic Large Break LOCA. However, because this was a CANDU:

- It resulted in a manageable **Small Break LOCA**. The heavy water poured into the reactor vault but was entirely contained.
- The outer calandria tube *did not fail*. The massive pool of cold moderator water absorbed the heat, acting as a passive heat sink and preventing any fuel melting.
- The reactor was safely shut down with **zero radioactive release** to the public or injuries to workers.

The Industry Fix & Restart: Following this incident, the industry realized standard Zircaloy-2 was too susceptible to hydride embrittlement. Both Unit 1 and Unit 2 underwent a massive, first-of-its-kind retubing effort to replace every single pressure tube. This unprecedented maintenance kept the reactors offline for years, with Unit 1 finally restarting in September 1987 and Unit 2 in October 1988. Modern CANDUs now strictly use a stronger **Zirconium-2.5% Niobium** alloy for their inner pressure tubes, alongside re-designed, tighter-fitting garter springs.

3 The Boiling Water Reactor (BWR)

The BWR (approx. 15% of global fleet) utilizes a **Direct Cycle**: light water boils directly in the core, and that same steam drives the turbine.

3.1 Circulation and Steam Separation

Contrary to the simple "pot on a stove" model, the water in a BWR is vigorously recirculated.

1. **The Recirculation Ratio:** Only a small fraction (approx. 14% by mass) of the water entering the core actually turns to steam in a single pass. The rest remains liquid.
2. **Steam Separation:** The two-phase mixture (liquid + vapor) exits the core and passes through **Steam Separators** (cyclones) and **Dryers** (chevron vanes) at the top of the vessel to remove moisture droplets.
 - *Steam Path:* Dry steam exits the vessel and goes directly to the High Pressure Turbine.

- *Liquid Path:* The removed water flows down the "downcomer" (annulus between core shroud and vessel wall), mixes with fresh feedwater, and is pumped back up through the core.

3.2 Recirculation Mechanisms

3.2.1 Forced Circulation: The Jet Pump System

Most operating BWRs (e.g., BWR/4, BWR/5) use **Jet Pumps** located inside the vessel annulus to drive this flow.

- **The Mechanism:** This is very similar to a residential "deep well" ejector pump.
- **Operation:** External recirculation pumps pull a small amount of water out of the vessel (the "driving flow") and shoot it back in at high velocity through a nozzle. This high-velocity jet entrains the surrounding water (the "suction flow") and drags it into the core.
- **Advantage:** You only need to pipe a fraction of the total core flow outside the vessel, significantly reducing the risk/size of a Loss of Coolant Accident (LOCA).
- **Control Strategy:** Reactor power is controlled by changing the speed of the external pumps.

Faster Flow $\rightarrow$ Sweeps bubbles out $\rightarrow$ Void fraction $\downarrow \rightarrow$ Moderation $\uparrow \rightarrow$ **Power Rises**

3.2.2 Natural Circulation (ESBWR)

Modern designs (like the GE Hitachi ESBWR) eliminate the pumps entirely to create a "passive" safety system.

- **Driving Force:** Natural Convection. A tall "chimney" section is added above the core. The density difference between the cold water in the downcomer and the hot steam/water mix in the chimney drives the flow.
- **Control:** Since we cannot adjust pump speed, power is controlled via Fine Motion Control Rod Drives (FMCRD) and by adjusting the temperature of the returning feedwater (feedwater inlet subcooling).

3.3 Operational Challenges

1. **Radioactive Turbine (N-16):** Because the primary coolant goes to the turbine, the steam contains Nitrogen-16, produced by fast neutron capture in oxygen:



^{16}N emits very high-energy gammas (6 MeV and 7 MeV).

- *Consequence:* The turbine hall is a high-radiation zone during operation and must be heavily shielded. However, because $T_{1/2} = 7.13$ seconds, the radiation levels drop to safe limits within minutes of shutdown.
2. **No Chemical Shim:** We cannot dissolve Boric Acid in the coolant for reactivity control because it would plate out on the fuel and turbine blades when the water boils. Reactivity is managed solely by flow control and control rods (plus burnable poisons like Gadolinium in the fuel).

3. **Bottom-Entry Control Rods:** The top of the reactor vessel is crowded with steam dryers. Therefore, control blades (cruciform shape) enter from the bottom.
 - *Safety Issue:* Gravity does not help scram the reactor. Rods must be driven in hydraulically using high-pressure accumulators.

4 BWR Containment Strategy and Fukushima

Unlike the massive "Large Dry" domes of PWRs, BWRs rely on a thermodynamic trick called **Pressure Suppression** to manage a LOCA. This allows for a much physically smaller containment structure.

4.1 The Mark I Containment (Pressure Suppression)

The original GE design (used at Fukushima Daiichi Units 1-5) is the Mark I, famous for its "lightbulb and torus" geometry.

- **Drywell (The Lightbulb):** A steel pressure vessel immediately surrounding the reactor. If a primary pipe breaks, expanding steam fills this confined space.
- **Wetwell (The Torus):** A massive donut-shaped tube running around the base of the reactor, half-filled with millions of gallons of cold water (the Suppression Pool).
- **The Mechanism:** Massive vent pipes connect the Drywell to the Wetwell. During a LOCA, the expanding steam is forced down the vents and bubbled through the cold water of the Suppression Pool, instantly condensing it. This rapidly quenches the steam, drastically reducing the peak accident pressure.

4.2 The Fukushima Daiichi Accident (2011)

The accident at Fukushima Daiichi was fundamentally a prolonged Station Blackout (SBO) leading to a complete loss of the ultimate heat sink.

1. **The Earthquake and SCRAM:** On March 11, a magnitude 9.0 earthquake struck. The reactors successfully SCRAMmed. Off-site power was lost, but emergency diesel generators started automatically to provide cooling flow to remove decay heat.
2. **The Tsunami and SBO:** 45 minutes later, a 14-meter tsunami overtopped the 5.7-meter seawall. It violently flooded the basements where the diesel generators and DC batteries were housed, causing a total Station Blackout.
3. **Loss of Cooling:** Without AC or DC power to control valves, the emergency core cooling systems (like the steam-driven RCIC system) eventually failed. The primary coolant in Units 1, 2, and 3 boiled away.
4. **Core Melt and Hydrogen Generation:** The uncovered fuel heated up, initiating the highly exothermic Zircaloy-steam reaction. This generated massive amounts of explosive hydrogen gas and led to core meltdowns in all three units.

5. **Containment Venting and Explosions:** To prevent the primary containment (Drywell) from rupturing due to overpressure, operators vented the gases. Unfortunately, the vented hydrogen accumulated in the upper floors of the secondary containment (the square reactor buildings).
6. **The Result:** The hydrogen detonated, blowing the roofs off Units 1, 3, and 4 (Unit 4 had shared piping with Unit 3). This released significant radioactive material into the environment, contaminating the region and profoundly altering global nuclear safety paradigms.

4.3 Post-Fukushima Mitigation and Advanced BWRs

Following the accident, the global nuclear industry fundamentally changed how it handles Station Blackouts to ensure later models and existing plants are not vulnerable to the same sequence of events.

- **The FLEX Strategy (Existing Fleet):** For currently operating reactors, the U.S. NRC mandated "Diverse and Flexible Coping Strategies" (FLEX). Plants must now store portable, diesel-driven pumps and generators in hardened, decentralized bunkers on-site, with pre-installed standardized hookups to inject water even if all permanently installed electrical systems fail.
- **Advanced Boiling Water Reactor (ABWR):** This Generation III design is an evolutionary step. While still "active" (requiring power), it utilizes internal recirculation pumps (eliminating large external loops) and includes an entirely independent, hardened gas-turbine generator in addition to multiple waterproofed diesel generators to survive prolonged blackouts.
 - *Commercial Deployment:* The ABWR is the only advanced BWR to actually be built. Four units successfully entered commercial operation in Japan in the 1990s and 2000s (prior to the post-Fukushima nationwide shutdown). Planned U.S. builds were ultimately cancelled due to the shifting economics of cheap natural gas.
- **The ESBWR and Passive Safety (The SBO Solution):** The GE-Hitachi Economic Simplified Boiling Water Reactor (ESBWR) was designed specifically to survive a Fukushima-style blackout. It is a true "walk-away safe" design.
 - *Gravity-Driven Cooling System (GDCCS):* Instead of relying on AC-powered pumps, massive pools of water are stored directly above the reactor inside containment. If power is lost, valves automatically open, and gravity drops the water into the core.
 - *Isolation Condensers:* Heat exchangers located in high-elevation pools naturally draw steam from the reactor, condense it, and return it via gravity without any electrical power.
 - *The 72-Hour Rule:* The ESBWR is licensed to safely cool itself during a total Station Blackout for a minimum of **72 hours** (and practically up to 7 days) without any AC power or operator intervention.
 - *Commercial Fate:* Despite receiving full NRC design certification in 2014, a full-scale ESBWR was never constructed. However, its passive cooling technology was directly scaled down to create the GE-Hitachi BWRX-300 Small Modular Reactor (SMR), which is currently under construction in Ontario.

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